Back **Asymmetric Cryptography Overview**
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Asymmetric Cryptography Overview

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1. Given any plaintext *p*, a cipher supporting asymmetric cryptography with an encryption function (*Enc*) and the corresponding decryption function (*Dec*), and the public-private key pair (*Ki*,*ki*) for any user *i*, which of the followings are true for a cipher that can be used for both message confidentiality and source integrity/signature, e.g., RSA cipher? Select all that applies. **4 / 4 points**

- [ ] Dec(k1,Enc(k1,p))=p
- [x] Dec(K2,Enc(k2,p))=p

  Correct
  See Asymmetric Encryption for Message Confidentiality lesson.
- [x] Dec(k1,Enc(K1,p))=p

  Correct
  See Asymmetric Encryption for Message Confidentiality lesson.
- [ ] Dec(K1,Enc(k2,p))=p

2. Which of the followings are true about asymmetric cryptography? Check all that applies. **4 / 4 points**

- [ ] Asymmetric cryptography is also called private-key cryptography.
- [x] Key distribution and management should be addressed when using asymmetric cryptography.

  Correct
  See the What is Asymmetric Cryptography lesson.
- [ ] Asymmetric cryptography supersedes and generalizes symmetric cryptography.
- [ ] Given the same key length, asymmetric cryptographic scheme is more secure than symmetric cryptographic scheme.

3. Which of the followings are **false** for asymmetric cipher requirements? Select all that applies. **4 / 4 points**

- [ ] It is computationally easy for any user to generate his/her own public-private key pair.
- [ ] The encryption and the decryption computations are easy only with the key that is being used.
- [x] Both the public key and the private key should remain secret against an attacker.

  Correct
  See Cipher Requirements and Trapdoor One-Way Function lesson.
- [ ] It is computationally infeasible for an attacker to derive the private key from a public key.
- [ ] It is computationally infeasible from an attacker to derive the plaintext from the public key and the ciphertext.
- [x] Both the sender and the receiver can use the same private key for encryption and decryption.

  Correct
  See Cipher Requirements and Trapdoor One-Way Function lesson.

4. Suppose f is a trapdoor one-way function designed to be used with the key, k. Which of the followings are computationally easy? **4 / 4 points**

- [x] Solving f(x) if the input and k are known

  Correct
  See Cipher Requirements and Trapdoor One-Way Function lesson.
- [x] Solving the inverse of f if the input to the f-inverse and k are known

  Correct
  See Cipher Requirements and Trapdoor One-Way Function lesson.
- [ ] Solving the inverse of f if the input to the f-inverse is known
- [ ] Finding k if the input and the corresponding output of f are known

5. Which of the followings does the RSA algorithm support? Select all that apply. **3 / 3 points**

- [x] Key exchange

  Correct
  See Digital Signature for Authentication lecture.
- [x] Encryption/decryption

  Correct
  See Digital Signature for Authentication lecture.
- [x] Digital signature

  Correct
  See Digital Signature for Authentication lecture.

6. Which of the followings does Diffie-Hellman Key Exchange support: encryption/decryption, digital sig- natures, key exchange? Select all that apply. **3 / 3 points**

- [x] Key exchange

  Correct
  See Digital Signature for Authentication lecture.
- [ ] Encryption/decryption
- [ ] Digital signature